

Mahmoud Alyosify

AI Instructor & Technical Trainer | Curriculum Design, Applied ML Teaching & Technical Content | MSc Artificial Intelligence, Queen's University (Canada)

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🌐 [linkedin.com/in/mahmoudalyosify](https://www.linkedin.com/in/mahmoudalyosify) | 🐙 github.com/MahmoudAlyosify | 📁 Portfolio | 🤖 Hugging Face | 🏆 Kaggle

PROFESSIONAL SUMMARY

AI Instructor and Technical Trainer with **5+ years** teaching computer science and machine learning to **10,000+ learners** across **700K+ YouTube views** — and, crucially, a practitioner who still builds. Designed and delivered a **90-hour hands-on ML program** at Egypt's **National Telecommunication Institute**, and authored Arabic-language curricula in **discrete mathematics, algorithms, probability, statistics and graph data mining** for a region under-served by technical content in its own language. Currently completing an **MSc in Artificial Intelligence at Queen's University, Canada** (expected **Nov 2026**) on a **Digilians Presidential Scholarship**, with active research and engineering work across LLMs, computer vision and reinforcement learning that keeps teaching material grounded in current practice. **1st Place, Science & Technology Content Competition.**

TEACHING & TRAINING EXPERIENCE

Machine Learning Instructor Jul 2025 – Sep 2025
National Telecommunication Institute (NTI) Cairo, Egypt

- Designed and delivered a **90-hour** hands-on machine learning program for engineering and computer-science students at Egypt's national ICT training institute.
- Covered supervised learning, **PCA** and neural networks, structuring every module around practical lab sessions rather than lecture-only delivery.
- Guided learners through building complete end-to-end ML solutions on real datasets, and adapted explanations for a technically mixed cohort.

Online Instructor — AI, Algorithms & Data Science 2020 – Present
Udemy & YouTube Remote

- Built and taught Arabic-language technical curricula in **discrete mathematics, algorithms, probability, statistics and graph data mining**, reaching **10,000+ learners** and **700K+ YouTube views**.
- Authored courses end to end — syllabus design, lesson scripting, recording, editing and ongoing learner support — sustaining high ratings across cohorts.
- Addressed a genuine gap: rigorous CS fundamentals taught in Arabic, for students who lose material to the language barrier before they lose it to difficulty.

Ambassador During undergraduate study
Information Technology Institute (ITI), Assiut Assiut, Egypt

- Represented ITI on campus, promoting technical training pathways and guiding students toward relevant programs.

EDUCATION

MSc in Artificial Intelligence Sep 2025 – Nov 2026
Queen's University, Kingston, Ontario, Canada Expected Nov 2026

- Digilians Presidential Scholarship** — competitive national AI graduate fellowship.
- Master's Research Project (CISC 898):** *Verbosity-Aware Decoding for Token-Efficient Language Model Inference* — Supervisor: Dr. Rahatara Ferdousi.
- Coursework: **Large Language Models, Generative AI, Deep Learning, Reinforcement Learning, Cloud Computing, End-to-End ML Pipelines.**

B.Sc. Computer & Information Science (Bioinformatics) Sep 2019 – Jul 2023
Assiut University, Egypt GPA: 3.53 / 4.00

- Graduation Project: **Vitalism Solution** — **Grade A+; Semi-Finalist, Microsoft Imagine Cup 2023.**

TEACHING & TECHNICAL SKILLS

Teaching & Delivery	Curriculum & Syllabus Design, Hands-On Lab Development, Lecture Delivery, Online Course Production, Technical Content Creation, Video Production & Editing, Learner Support, Public Speaking, Bilingual (Arabic / English) Instruction
Subjects Taught	Machine Learning, Neural Networks, PCA & Dimensionality Reduction, Supervised Learning, Discrete Mathematics, Algorithms, Probability & Statistics, Graph Data Mining, Programming Fundamentals
AI & ML	PyTorch, TensorFlow, Scikit-learn, Hugging Face, Large Language Models, Fine-Tuning (LoRA, QLoRA), RAG, Computer Vision, Reinforcement Learning (PPO, DQN), Multi-Agent Systems
Engineering	Python, SQL, C++, C#, C, JavaScript, R, MATLAB, AWS, Docker, FastAPI, Streamlit, ONNX, Git/GitHub, Linux

PRACTITIONER WORK BEHIND THE TEACHING

- **Verbosity-Aware Decoding** — MSc research on training-free token reduction at decoding time, validated across three model families on MT-Bench, XSum and SQuAD-v2 with latency and GPU-energy measurement.
- **HORUS Sentinel** — autonomous multi-agent OSINT platform with a knowledge graph and self-hosted fine-tuned LLM.
- **SimCLR-Vision-SSL** — self-supervised vision pipeline, **42-experiment** augmentation sweep, **84.30% top-1**, ONNX/FAISS deployment.
- **RL-Cloud-Autoscaler** — custom Gymnasium environment with PPO and DQN agents optimising a cost-latency trade-off.
- **Horus-OSINT** — fine-tuned **Meta-Llama-3-8B** on **159,826** records, deployed on AWS.
- **Also:** Automated Multimodal Agent · SCRAP · AudioShield · Vitalism Solution (rPPG) · Human vs. Agentic Pull Requests.

ADDITIONAL PROFESSIONAL EXPERIENCE

- AI in Cybersecurity Intern**
e& Egypt — via NTI (On-the-Job Training)

Jun 2025 – Jul 2025
Cairo, Egypt

 - Applied AI-driven security tooling across threat intelligence, digital forensics and enterprise security analysis.
- Systems Engineer (Military Service)**
National Company for Roads Building & Development

Feb 2024 – Mar 2025
Cairo, Egypt

 - Supported backend systems for national toll infrastructure serving **50%+ of Egyptian vehicles**; co-developed an internal data-management application in **C#, .NET, SQL Server**.

AWARDS, CERTIFICATIONS & LANGUAGES

- 1st Place** — Science & Technology Content Competition

Ideal Student Award — Assiut University, 2022–2023
- Digilians Presidential Scholarship** — Queen’s University

1st Place — Smart Cities Hackathon 2022
- Machine Learning Specialization** — DeepLearning.AI

AWS Academy — ML Foundations, ML for NLP, Data Engineering
- HCIP – AI (Huawei)**

Communication Skills: Star Presenter on Panels
- **Languages:** Arabic (Native) · English (Professional Working Proficiency) · French (Basic).
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